

How to Assign an Access Password Before Using Protected Mode

Protected Mode requires the target tag to already have a valid 32-bit access password. Assign one by writing 8 hexadecimal characters to the Gen2 **RESERVED** bank at **wordPointer: 2**.

Words	Contents
0-1	Kill password
2-3	Access password <- write here

1. Assign the password

REST - PUT /cloud/mode

```
{
  "type": "CUSTOM",
  "antennas": [1],
  "transmitPower": 30,
  "filter": {
    "value": "e28011b0a5050076c4d7530a",
    "match": "prefix",
    "operation": "include"
  },
  "accesses": [
    {
      "type": "WRITE",
      "config": {
        "membank": "RESERVED",
        "wordPointer": 2,
        "data": "A1B2C3D4"
      }
    }
  ]
}
```

MQTT - set_mode

```
{
  "command": "set_mode",
  "payload": {
    "type": "CUSTOM",
    "antennas": [1],
    "transmitPower": 30,
    "filter": {
      "value": "e28011b0a5050076c4d7530a",
      "match": "prefix",
      "operation": "include"
    },
    "accesses": [
      {
        "type": "WRITE",
        "config": {
          "membank": "RESERVED",
          "wordPointer": 2,
          "data": "A1B2C3D4"
        }
      }
    ]
  }
}
```

Then run an inventory - the write executes when the tag is singulated, not when the command returns:

Action	REST	MQTT
Start	PUT /cloud/start with {}	{"command": "start", "payload": {}}
Stop	PUT /cloud/stop	{"command": "stop", "payload": {}}

Result in the tag data stream:

```
{"data": {"idHex": "e28011b0a5050076c4d7530a", "accessResults": ["SUCCESS"]}}
```

Include the **filter**. Without it the password is written to **every tag in the field**.

2. Use it with Protected Mode

REST - PUT /cloud/impinjGen2X

```
{
  "tagProtect": {
    "action": "enableTagProtection",
    "password": "A1B2C3D4",
    "tagID": "e28011b0a5050076c4d7530a"
  }
}
```

MQTT - set_impinjGen2X

```
{
  "command": "set_impinjGen2X",
  "payload": {
    "tagProtect": {
      "action": "enableTagProtection",
      "password": "A1B2C3D4",
      "tagID": "e28011b0a5050076c4d7530a"
    }
  }
}
```

Apply it on the next start:

REST	MQTT
PUT /cloud/start with {"applyImpinjGen2X": true}	{"command": "start", "payload": {"applyImpinjGen2X": true}}

Password rules

- Exactly **8 hex characters** (0-9, A-F) - 32 bits
- Case-insensitive: **a1b2c3d4** = **A1B2C3D4**
- **Do not use `00000000`** - that means "no password" on a Gen2 tag, so it cannot protect anything. The API accepts it without warning.

Commands used

Purpose	REST	MQTT
Write the password	PUT /cloud/mode	set_mode
Start inventory	PUT /cloud/start	start
Stop inventory	PUT /cloud/stop	stop
Configure Protected Mode	PUT /cloud/impinjGen2X	set_impinjGen2X

Verified on FXR60 5.0.7 and FXR90 5.0.4 with Impinj Monza 4i tags.